AFTERSHOCK

Perception-driven, safety-gated disaster-response robotics | AgentVerse 2026 | Technical summary

USER & PROBLEM

Earthquake-damaged buildings expose rescuers to unstable debris, secondary collapse and restricted access. Early survivor localisation is urgent, yet teams may lack reliable information about interior hazards and casualty positions. Rescue teams need to understand hazards in damaged buildings before sending people or heavy equipment in. AFTERSHOCK simulates a Scout mapping a collapse**d** building and a Responder delivering supplies on routes checked in physics. A scripted aftershock tests adaptation to change.

WHY A ROBOTIC AGENT?

Scout-first reconnaissance can gather hazard and casualty evidence before personnel enter; a supervised Responder can test routes and deliver limited supplies. The aim is to reduce unnecessary human exposure and improve rescue decisions—not replace trained rescuers or claim proven survival benefits.

KEY FEATURES

Observe: LiDAR/depth mapping, thermal + acoustic + optional vision casualty cues, observed/inferred/unknown map cells.

Decide: confidence-aware exploration, hazard estimates, local A* route planning and physics verification.

Adapt: an aftershock invalidates the old route; the Scout re-scans and the planner re-routes.

Act & explain: simulated Responder delivers supplies, operator approves risky actions, browser shows evidence and reasoning, voice/typed interaction and an auditable handover report.

SYSTEM ARCHITECTURE

MuJoCo scene + Scout sensors **LiDAR / depth / thermal / audio**	**Perception** **Map, casualty confidence, hazard / motion zones**	**Planner + verifier** **A* route, MuJoCo physics, human gate**
Responder simulator **Move / grip / supply placement**	Async topic bus + FastAPI Validated events; WebSocket /ws	Operator console + voice 3D map, approvals, reports, speech

MODEL PLACEMENT, HARDWARE & LATENCY

Model / function	Execution location and hardware	Latency / budget
Local A* + safety gate	Local Python; laptop CPU, 8 GB RAM	Measured <50 ms per planning/gate step
Gemma 4 E2B (offline goals)	Ollama Q4_K_M; 8 GB RAM laptop, no GPU; 4.1 GB resident	Measured 1.8–2.8 s warm; ~19 s cold; 4 s step target
Gemini 4 Argon (reasoning / triage)	Cloud API; local fallback when unavailable	Not measured; not on critical motion path
Gemini 3.8 Flash (vision / report)	Cloud API; vision triggered at most 1/s	End-to-end latency not measured
Gemini 3.8 Live / Flash TTS; 3.5 Live Translate	Cloud API for speech / translation; text fallback	End-to-end latency not measured

CRITICAL FAILURE MODE & SAFETY

Failure mode: an aftershock moves debris after a route has been approved, making a previously safe corridor unsafe.

Mitigation: sensed change revokes the route permit and stops/holds simulated motion; new Scout measurements, hazard exclusion zones and a fresh physics check are required before movement resumes. Unknown/inferred cells cannot be treated as confirmed safe. High-risk actions need explicit human approval; denial or a 60 s timeout leaves the Responder idle. Approval cannot override failed verification. Emergency calls are prepared, not automatically placed.

LIMITS & EVIDENCE

Simulation-based proof of concept: sensor streams, casualties and vital signs are simulated; structural risk is an estimate, not engineering certification. Mapping may miss occluded or elevated debris; routes require robot-footprint and real-world validation. Cloud model latencies have not been benchmarked, and live services may be unavailable. Hardware (Jetson-class Scout/Responder, cameras, LiDAR, watchdog and e-stop) is a future design, not deployed or safety-certified.

TRACK FIT

Shows sensor-driven action and visible replanning after change.

Video: sensing beside action, one unscripted change, one take, up to one minute.